

Homogenization of Elliptic Systems with Lower Order Terms

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Background

In "Uniform regularity estimates in homogenization theory of elliptic system with lower order terms" Q. Xu studied the homogenization problems of Elliptic operator with lower order terms in $\mathbb{R}^d$ with $d \geq 3$. That is, for the elliptic operator

$$\mathcal{L}_\varepsilon = -\operatorname{div}[A(x/\varepsilon)\nabla + V(x/\varepsilon)] + B(x/\varepsilon)\nabla + c(x/\varepsilon) + \lambda I$$

Q. Xu obtained uniform regularity estimates independent of $\varepsilon > 0$.

We considered the case for $d = 2$. Precisely speaking, we mainly consider the $W^{1,p}$ and Hölder estimates for such operators.

The main tools is some iterations arguments and Green functions.

Some conditions of coefficients of $\mathcal{L}_\varepsilon$

(1) Ellipticity condition: $\exists \mu > 0$, s.t. $\forall \xi = (\xi_i^\alpha) \in \mathbb{R}^{m \times d}$

$$\mu |\xi|^2 \leq a_{ij}^{\alpha\beta}(y) \xi_i^\alpha \xi_j^\beta \leq \mu^{-1} |\xi|^2.$$

(2) Periodicity condition: $\forall y \in \mathbb{R}^d, z \in \mathbb{Z}^d$

$$\begin{aligned} A(y+z) &= A(y), V(y+z) = V(y), \\ B(y+z) &= B(y), c(y+z) = c(y). \end{aligned}$$

(3) Boundedness condition: $\exists \kappa > 0$, s.t.

$$\max \left\{ \|V\|_{L^\infty(\mathbb{R}^d)}, \|B\|_{L^\infty(\mathbb{R}^d)}, \|c\|_{L^\infty(\mathbb{R}^d)} \right\} \leq \kappa.$$

(4) VMO condition: $\exists \omega(t) : [0, \infty) \rightarrow [0, \infty)$ non decreasing, s.t.

$$\sup_{x \in \mathbb{R}^d, 0 < \rho < t} \oint_{B(x, \rho)} |A(y) - \oint_{B(x, \rho)} A| dy \leq \omega(t), \text{ and } \lim_{t \rightarrow 0} \omega(t) = 0,$$

Uniform $W^{1,p}$ estimates and Hölder estimates

Theorem (Uniform $W^{1,p}$ estimates)

$p > 2$, $q = 2p/(p+2)$; $p = 2$, $1 < q < \infty$; $1 < p < 2$, $q = 1$,
 $\lambda > 0$ sufficiently large, Ω bounded C^1 in $\mathbb{R}^2$. $\exists! u_\varepsilon$ s.t.

$$\mathcal{L}_\varepsilon(u_\varepsilon) = \operatorname{div}(f) + F \text{ in } \Omega, \quad u_\varepsilon = g \text{ on } \partial\Omega,$$

$$\|\nabla u_\varepsilon\|_{L^p(\Omega)} \lesssim \|F\|_{L^q(\Omega)} + \|f\|_{L^p(\Omega)} + \|g\|_{B^{1-\frac{1}{p},p}(\partial\Omega)}.$$

Theorem (Hölder estimates)

$2 < p < \infty$, $\sigma = 1 - 2/p$, $q = 2p/(p+2)$, $\lambda > 0$ sufficiently large,
 Ω bounded C^1 in $\mathbb{R}^2$. $\exists! u_\varepsilon$, s.t.

$$\mathcal{L}_\varepsilon(u_\varepsilon) = \operatorname{div}(f) + F \text{ in } \Omega, \quad u_\varepsilon = g \text{ on } \partial\Omega,$$

$$\|u_\varepsilon\|_{C^{0,\sigma}(\Omega)} \lesssim \|F\|_{L^q(\Omega)} + \|f\|_{L^p(\Omega)} + \|g\|_{C^{0,\sigma}(\partial\Omega)}.$$

Here, the Hölder estimates are Sharp. If $g \in C^{0,1}(\partial\Omega)$, the result is natural by using Sobolev embedding theorem.

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Existence and Homogenization

For operators $\mathcal{L}_\varepsilon$, we have the following results.

Lemma (Preparations for existence)

For $\lambda > 0$ sufficiently large, $\forall u, v \in W_0^{1,2}(\Omega; \mathbb{R}^m)$

$$|\langle \mathcal{L}_\varepsilon(u), v \rangle| \leq C \|u\|_{W_0^{1,2}(\Omega)} \|v\|_{W_0^{1,2}(\Omega)},$$

$$c_0 \|u\|_{W_0^{1,2}(\Omega)}^2 \leq \langle \mathcal{L}_\varepsilon(u), u \rangle,$$

Theorem (Existence and homogenization theorem)

If $\lambda > 0$ sufficiently large, the Dirichlet problem $\mathcal{L}_\varepsilon(u_\varepsilon) = F$ in Ω , $u_\varepsilon = g$ on $\partial\Omega$ has a unique weak solution $u_\varepsilon \in W^{1,2}(\Omega; \mathbb{R}^m)$, s.t.

$$\|u_\varepsilon\|_{W^{1,2}(\Omega)} \lesssim \|F\|_{W^{-1,2}(\Omega)} + \|g\|_{W^{1/2,2}(\partial\Omega)}.$$

When $\varepsilon \rightarrow 0$, $u_\varepsilon \rightarrow u_0$ in $W^{1,2}(\Omega; \mathbb{R}^m)$, $u_\varepsilon \rightarrow u_0$ in $L^2(\Omega; \mathbb{R}^m)$, where u_0 satisfies $\mathcal{L}_0(u) = 0$ in Ω , $u = g$ on $\partial\Omega$.

Caccioppoli's Inequality and Interpolation Inequalities

Lemma (Caccioppoli's inequality)

$q > 1$. $B(x_0, r) \subset B(x_0, 2r) \subset \Omega$, $0 < r \leq 1$, $x_0 \in \Omega$, then

$$\|\nabla u_\varepsilon\|_{L^2_{\text{avg}}(B)} \lesssim r^{-1} \|u_\varepsilon\|_{L^2_{\text{avg}}(2B)} + \|f\|_{L^2_{\text{avg}}(2B)} + r \|F\|_{L^q_{\text{avg}}(2B)},$$

where $\|u\|_{L^p_{\text{avg}}(E)} = (f_E |u|^p)^{1/p} = (\frac{1}{|E|} \int_E |u|^p)^{1/p}$ with $E \subset \mathbb{R}^2$.

Lemma (Interpolation inequalities)

$u \in W^{1,p}(\Omega; \mathbb{R}^m)$, $2 \leq p < \infty$, $\forall \delta > 0$, $\exists C_\delta > 0$ s.t.

$$\|u\|_{L^p(\Omega)} \leq \delta \|\nabla u\|_{L^p(\Omega)} + C_\delta \|u\|_{L^2(\Omega)}.$$

If $1 \leq p < \infty$, $q = pd/(p+d)$, then

$$\|u\|_{L^p(\Omega)} \lesssim \|\nabla u\|_{L^q(\Omega)} + \|u\|_{L^2(\Omega)}.$$

The index q differs from $d \geq 3$ and $d = 2$, since $H_0^1 \subset L^q$ for $q > 1$ when $d = 2$, instead of $H_0^1 \subset L^{\frac{2d}{d-2}}$, when $d \geq 3$.

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The Proof of $W^{1,p}$ Estimates

Theorem (Without lower order terms)

Let $L_\varepsilon = -\operatorname{div}(A(x/\varepsilon)\nabla)$. $p > 2$, $q = 2p/(p+2)$; $p = 2$, $1 < q < \infty$; $1 < p < 2$, $q = 1$, Ω bounded C^1 in $\mathbb{R}^2$.

$\exists! u_\varepsilon \in W^{1,p}(\Omega; \mathbb{R}^m)$ s.t. $L_\varepsilon(u_\varepsilon) = \operatorname{div}(f) + F$ in Ω , $u_\varepsilon = 0$ on $\partial\Omega$ and

$$\|\nabla u_\varepsilon\|_{L^p(\Omega)} \lesssim \|F\|_{L^q(\Omega)} + \|f\|_{L^p(\Omega)}$$

Lemma (First step for the proof of $W^{1,p}$ estimates)

$1 < p < \infty$, $\lambda > 0$ sufficiently large, Ω bounded C^1 in $\mathbb{R}^2$.

$\exists! u_\varepsilon \in W^{1,p}(\Omega; \mathbb{R}^m)$ s.t. $\mathcal{L}_\varepsilon(u_\varepsilon) = \operatorname{div}(f)$ in Ω , $u_\varepsilon = 0$ on $\partial\Omega$ and,

$$\|\nabla u_\varepsilon\|_{L^p(\Omega)} \lesssim \|f\|_{L^p(\Omega)}.$$

$p = 2$, energy estimates. $p > 2$, results for L_ε and iterations.

$p < 2$, duality arguments. This lemma gives the proof of $W^{1,p}$

Hölder Estimates and Green Functions

We use methods developed by S. Kim and coworkers in "Green's function for nondivergence elliptic operators in two dimensions", "The Green function for elliptic systems in two dimensions", as well as for the case $d \geq 3$ given by Q. Xu.

Theorem (Green functions)

λ sufficiently large, Ω bounded C^1 in $\mathbb{R}^2$. Then there exists unique Green functions $G_\varepsilon : \Omega \times \Omega \rightarrow \mathbb{R}^{m^2} \cup \{\infty\}$ s.t. $\forall u_\varepsilon$ solving $\mathcal{L}_\varepsilon(u_\varepsilon) = F$ in Ω , $u_\varepsilon = 0$ on $\partial\Omega$, where $F \in L^q(\Omega; \mathbb{R}^m)$, ($q > 1$), then

$$u_\varepsilon(x) = \int_\Omega G_\varepsilon(x, y) F(y) dy.$$

Moreover, for Green functions $G_\varepsilon^(x, y)$ for the adjoint operator of $\mathcal{L}_\varepsilon$, namely $\mathcal{L}_\varepsilon^*$, we have $G_\varepsilon^*(x, y) = G_\varepsilon(y, x)$.*

$\forall \sigma_1, \sigma_2, \sigma \in (0, 1)$,

$$G_\varepsilon(\cdot, y) \in BMO(\Omega) \text{ s.t. } \|G_\varepsilon(\cdot, y)\|_* \leq C, \quad \forall y \in \Omega,$$

Theorem (Continued from previous page)

$$|G_\varepsilon(x, y)| \lesssim \frac{[\delta(x)]^{\sigma_1}}{|x - y|^{\sigma_1}}, \delta(x) < \frac{1}{4}|x - y|,$$

$$|G_\varepsilon(x, y)| \lesssim \frac{[\delta(y)]^{\sigma_2}}{|x - y|^{\sigma_2}}, \delta(y) < \frac{1}{4}|x - y|,$$

$$|G_\varepsilon(x, y)| \lesssim \frac{[\delta(x)]^{\sigma_1} [\delta(y)]^{\sigma_2}}{|x - y|^{\sigma_1 + \sigma_2}}, \delta(x) < \frac{1}{4}|x - y| \text{ or } \delta(y) < \frac{1}{4}|x - y|,$$

$$|G_\varepsilon(x, y)| \lesssim C \left(1 + \ln \left(\frac{\text{diam}(\Omega)}{|x - y|} \right) \right),$$

$$\delta(x) \geq \frac{1}{4}|x - y| \text{ and } \delta(y) \geq \frac{1}{4}|x - y|,$$

$\delta(x) = \text{dist}(x, \partial\Omega)$ and $\text{diam}(\Omega)$ is the diameter of Ω .

An Important Lemma and Some Remarks

Such constructions of Green functions can also be used for non homogenization problems. That is the method in the thesis is universal for General elliptic operator in $\mathbb{R}^2$.

Lemma (An important lemma)

$\forall \sigma \in (0, 1)$, Ω bounded C^1 in $\mathbb{R}^2$, $G_\varepsilon(x, y)$ the Green functions, then

$$\int_{\Omega} |\nabla_y G_\varepsilon(x, y)| [\delta(y)]^{\sigma-1} dy \lesssim [\delta(x)]^\sigma.$$

This lemma is crucial to the proof of Hölder estimates. The proof of this lemma is different from the case for $d \geq 3$, even for the elliptic operator without lower order terms.

By using this lemma and standard arguments, we can complete the proof of the Sharp Hölder estimates.

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Some Conditions

In order to get the conclusion of results of eigenvalues and eigenfunctions, we need to give the following symmetry conditions about the coefficient matrix of $\mathcal{L}_\varepsilon$,

$$V(y) = B(y), B^* = B, A^* = A, c^* = c,$$

That is, for any $y \in \mathbb{R}^d$, $1 \leq i \leq d$ and $1 \leq \alpha, \beta \leq m$, we have, $V_i^{\alpha\beta}(y) = B_i^{\alpha\beta}(y) = B_i^{\beta\alpha}(y)$, $a_{ij}^{\alpha\beta}(y) = a_{ji}^{\beta\alpha}(y)$ and $c^{\alpha\beta}(y) = c^{\beta\alpha}(y)$.

In addition, in some cases, higher regularity requirements for coefficients are as follows

$$\max \left\{ \|A\|_{C^{0,\tau}(\mathbb{R}^d)}, \|V\|_{C^{0,\tau}(\mathbb{R}^d)}, \|B\|_{C^{0,\tau}(\mathbb{R}^d)} \right\} \leq \kappa.$$

Problems of eigenvalues and eigenfunctions

Obviously, for $\lambda \geq \lambda_0$ (λ_0 sufficiently large), $\mathcal{L}_\varepsilon$ is a positive operator. The symmetry conditions implies that it is self adjoint. For $f \in L^2(\Omega)$, define the unique weak solution of $\mathcal{L}_\varepsilon(u_\varepsilon) = f$ in $W_0^{1,2}(\Omega; \mathbb{R}^m)$ is $u_\varepsilon = T_\varepsilon^D(f)$, $\varepsilon \geq 0$. Then

$$\langle T_\varepsilon^D(f), f \rangle = \langle u_\varepsilon, f \rangle,$$

This gives $\|u_\varepsilon\|_{H_0^1(\Omega)} \leq C\|f\|_{L^2(\Omega)}$. Then T_ε^D is compact. For $\varepsilon \geq 0$, we can define

$$\theta_{\varepsilon,1} \geq \theta_{\varepsilon,2} \geq \dots \geq \theta_{\varepsilon,k} \geq \dots > 0$$

is the sequence of eigenvalues of T_ε^D in descending order. By the min-max principle, for $\varepsilon \geq 0$,

$$\theta_{\varepsilon,k} = \min_{\substack{f_1, f_2, \dots, f_{k-1} \\ \in L^2(\Omega; \mathbb{R}^m)}} \max_{\substack{\|f\|_{L^2(\Omega)}=1, \\ f \perp f_i, i=1, \dots, k-1}} \langle T_\varepsilon^D(f), f \rangle,$$

Problems of Eigenvalues and Eigenfunctions

Let $\{\phi_{\varepsilon,k}\}$ is an orthogonal normal basis of $L^2(\Omega; \mathbb{R}^m)$ such that $\phi_{\varepsilon,k}$ is the eigenfunction related to the eigenvalue $\theta_{\varepsilon,k}$. Let $V_{\varepsilon,0} = \{0\}$ and $V_{\varepsilon,k}$ be $L^2(\Omega; \mathbb{R}^m)$ the subspace generated by $\{\phi_{\varepsilon,i}\}_{i=1}^k$. Then,

$$\theta_{\varepsilon,k} = \max_{f \perp V_{\varepsilon,k-1}, \|f\|_{L^2(\Omega)}=1} \langle T_{\varepsilon}^D(f), f \rangle.$$

Let $\eta_{\varepsilon,k} = (\theta_{\varepsilon,k})^{-1}$, then $\{\eta_{\varepsilon,k}\}$ is a sequence of increasing eigenvalues of $\mathcal{L}_{\varepsilon}$ in Ω .

Projection Operators

$\lambda \geq \lambda_0$. Let $\eta \geq 1$ Then the projection operators $S_{\varepsilon,\eta}(f)$ are defined as follows

$$S_{\varepsilon,\eta}(f) = \sum_{\sqrt{\eta_{\varepsilon,k}} \in [\sqrt{\eta}, \sqrt{\eta}+1)} \langle \phi_{\varepsilon,k}, f \rangle \phi_{\varepsilon,k}(x),$$

where $\{\phi_{\varepsilon,k}\}$ is an orthogonal normal basis of $L^2(\Omega; \mathbb{R}^m)$, such that $\mathcal{L}_{\varepsilon}(\phi_{\varepsilon,k}) = \eta_k \phi_{\varepsilon,k}$ for any k . Note that

$$\mathcal{L}_{\varepsilon}(S_{\varepsilon,\eta}(f)) = \eta S_{\varepsilon,\eta}(f) + R_{\varepsilon,\eta}(f),$$

where

$$R_{\varepsilon,\eta}(f) = \sum_{\sqrt{\eta_{\varepsilon,k}} \in [\sqrt{\eta}, \sqrt{\eta}+1)} (\eta_{\varepsilon,k} - \eta) \langle \phi_{\varepsilon,k}, f \rangle \phi_{\varepsilon,k}(x).$$

Main Results

We generalize the results of eigenvalues and eigenfunctions for L_ε given in "Estimates of eigenvalues and eigenfunctions in periodic homogenization" to the operator $\mathcal{L}_\varepsilon$.

Theorem (Convergence of eigenvalues)

$$|\eta_{\varepsilon,k} - \eta_{0,k}| \lesssim \varepsilon(\eta_{0,k})^{3/2}.$$

Theorem (Estimates for projection operators)

Let $u_\varepsilon = S_{\varepsilon,\eta}(f)$, where $f \in L^2(\Omega; \mathbb{R}^m)$ and $\|f\|_{L^2(\Omega)} = 1$. Then

$$\int_{\partial\Omega} |\nabla u_\varepsilon|^2 d\sigma \lesssim \begin{cases} \eta(1 + \varepsilon^{-1}) & \text{if } \varepsilon^2\eta \geq 1, \\ \eta(1 + \varepsilon\eta) & \text{if } \varepsilon^2\eta < 1. \end{cases}$$

In fact, we can get the estimates of eigenfunctions by taking η as a specific eigenvalue.

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Theorem (Estimates of Convergence rates)

$\lambda > 0$ sufficiently large. Let u_ε be the unique weak solution of the Dirichlet problem $\mathcal{L}_\varepsilon(u_\varepsilon) = f$ in Ω and $u_\varepsilon = 0$ on $\partial\Omega$. Then

$$\|u_\varepsilon - \Phi_{\varepsilon,0}u_0 - \left(\Phi_{\varepsilon,k}^\beta - P_k^\beta\right) \frac{\partial u_0^\beta}{\partial x_k}\|_{H_0^1(\Omega)} \lesssim \varepsilon \|f\|_{L^2(\Omega)}.$$

Moreover,

$$\|u_\varepsilon - u_0\|_{L^q(\Omega)} \lesssim \varepsilon \|f\|_{L^p(\Omega)}$$

for any $f \in L^p(\Omega; \mathbb{R}^m)$, where $q = \frac{pd}{d-p}$, if $1 < p < d$, $q = \infty$, if $p > d$.

Here $\Phi_{\varepsilon,k} = (\Phi_{\varepsilon,k}^\beta)$, $0 \leq k \leq d$, are the Dirichlet correctors for $\mathcal{L}_\varepsilon$ s.t.

$$L_\varepsilon(\Phi_{\varepsilon,0}) = \operatorname{div}(V_\varepsilon) \text{ in } \Omega, \quad \Phi_{\varepsilon,0} = I \text{ on } \partial\Omega$$

and

$$L_\varepsilon(\Phi_{\varepsilon,k}^\beta) = 0 \text{ in } \Omega, \quad \Phi_{\varepsilon,k}^\beta = P_k^\beta \text{ on } \partial\Omega,$$

where $1 \leq k \leq d$, where $V_\varepsilon = V(x/\varepsilon)$ and $P_k^\beta = x_k e^\beta$.

Estimates of Correctors and the Idea of proof

Let Ω is a bounded $C^{1,\eta}$ domain. If $\eta \in [\tau, 1)$, then, we can get the estimates of correctors as follows.

Lemma (Estimates of Correctors I)

$\forall \sigma \in (0, 1)$, then

$$\|\Phi_{\varepsilon,0} - I\|_{L^\infty(\Omega)} \lesssim \varepsilon, \|\Phi_{\varepsilon,0} - I\|_{C^{0,\sigma}(\Omega)} \lesssim \varepsilon^{1-\sigma}, \|\nabla \Phi_{\varepsilon,0}\|_{L^\infty(\Omega)} \lesssim 1,$$

Lemma (Estimates of Correctors II)

$$\|\Phi_{\varepsilon,k}^\beta - P_k^\beta\|_{L^\infty(\Omega)} \lesssim \varepsilon \text{ and } \|\nabla \Phi_{\varepsilon,k}\|_{L^\infty(\Omega)} \lesssim 1,$$

- For the problem of the convergence rates of eigenvalues, we use estimates of convergence rates, combined with estimates I, II.
- For estimates of projection operators, we established new Rellich identity for the operator $\mathcal{L}_\varepsilon$.

Thank you for listening!